PROJECT CAPSTONE

HOUSING PRICE PREDICTION- CALIFORNIA

Step 1:

This step imports libraries for data manipulation, visualization, preprocessing, modeling and evaluation.

Step2:

Perform EDA (Exploratory Data Analysis)- to analyse the data, visualise and understand

We loaded the dataset and inspected the first few rows to verify column names and data formats. Confirm which column will be used as the target (house price).

Step 3:

Perform descriptive statistics and plot graphs, find missing values

Calculated missing value counts and target statistics (count, mean, std, percentiles), and target histogram

Describe missing data patterns (which columns have missing values and how many).

Describe target distribution: skewness

Step 4:

Preprocessing and encode categorical variables

Location is a strong predictor; clustering lat/long into regions reduces cardinality and keeps locality signal.

IQR capping reduces influence of extreme outliers without removing rows.

Median imputation is robust to outliers.

Scaling standardizes features (important for linear models).

One-hot encoding converts categorical values into binary columns.

Use an 80/20 split to estimate final performance on held-out data. random_state ensures reproducible results.

Step 5:

Training data set, use linear models and calculate RSME

Calculations:

- CV RMSE: average RMSE from cross-validation on the training set.
- Test RMSE: RMSE on hold-out test set.
- Test R2: R^2 on test set.
- Test MAE: Mean Absolute Error on test set

Tree importances indicate which features the model used most to reduce prediction error (no direction). Top features are the ones to prioritize in feature engineering or data collection

Definitions

- **RMSE (Root Mean Squared Error):**
  RMSE = sqrt(mean((y_true − y_pred)²)). It measures the typical magnitude of prediction errors (same units as the target). Lower values indicate better performance.
- **MAE (Mean Absolute Error):**
  MAE = mean(|y_true − y_pred|). It gives the average absolute error and is less sensitive to outliers than RMSE.
- **R^2 (Coefficient of determination):**
  R^2 = 1 − SS_res / SS_tot where SS_res is the sum of squared residuals and SS_tot is the total sum of squares. It measures the fraction of variance explained by the model (1 means perfect, 0 means no improvement over predicting the mean

Model Performance Results

Linear Regression

- **CV RMSE:** 66,484.73
- **Test RMSE:** 70,384.41
- **Test R^2:** 0.6220

Ridge Regression

- **CV RMSE:** 66,477.16

- **Test RMSE:** 70,387.21
- **Test R^2:** 0.6219

Random Forest

- **CV RMSE:** 50,664.35
- **Test RMSE:** 50,047.80
- **Test R^2:** 0.8089

	LINEAR REGRESSION	RIDGE REGRESSION	RANDOM FOREST
CV RSME	66,484.73	66,477.16	50,664.35
Test RSME	70,384.41	70,387.21	50,047.80
Test R^2	0.6220	0.6219	0.8089

Interpretation: RMSE & R^2 Model Evaluation

1. Linear Regression — Interpretation

Performance:

- CV RMSE: **66,484.73**
- Test RMSE: **70,384.41**
- Test R^2: **0.6220**

Interpretation:
Linear Regression shows a test RMSE of around **$70k**, meaning its predictions deviate from actual house prices by roughly $70,000 on average.
The R^2 value of **0.6220** means the model is able to explain **62.20%** of the variance in house prices.

This is moderate performance, indicating the model captures only basic linear relationships. The Test RMSE being larger than CV RMSE shows slight underfitting — the model struggles with complex patterns in housing data.

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2. Ridge Regression — Interpretation

Performance:

- CV RMSE: **66,477.16**

- Test RMSE: **70,387.21**
- Test R^2: **0.6219**

Interpretation:
Ridge Regression performs almost identically to Linear Regression. The RMSE and R^2 values differ by only negligible margins.

This indicates that multicollinearity was not a major issue in the dataset, because the L2 regularization didn't significantly improve model accuracy.

Overall, Ridge is also underfitting and unable to capture important non-linear interactions present in the housing dataset.

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3. Random Forest — Interpretation (Best Model)

Performance:

- CV RMSE: **50,664.35**
- Test RMSE: **50,047.80**
- Test R^2: **0.8089**

Interpretation:
Random Forest gives a Test RMSE of **~$50k**, which is **far lower** than both Linear and Ridge. Its R^2 of **0.8089** shows that it explains **80.89%** of the variation in house prices — a significant improvement over linear models.

The closeness of CV RMSE and Test RMSE indicates the model generalizes extremely well with minimal overfitting.
Its ability to model non-linear relationships and feature interactions makes it much more suitable for this dataset.

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4. Final Model Comparison & Best Model Selection

Model	Test RMSE	Test R^2	Interpretation
Linear Regression	70,384	0.62	Weak. Cannot capture non-linear patterns.
Ridge Regression	70,387	0.6219	Same as Linear Reg. Regularization did not help

Model	Test RMSE	Test R^2	Interpretation
Random Forest	**50,047**	**0.8089**	**Best model**. High accuracy, strong R^2

Conclusion:

Random Forest is clearly the superior model with:

- **Lowest RMSE (~$50k)**
- **Highest R^2 (0.8089)**
- **Best generalization**
- **Ability to capture non-linear relationships**

Therefore, **Random Forest is selected as the final model** for the California House Price Prediction project.

Based on the evaluation metrics, Random Forest is the best-performing model. It achieves a Test RMSE of **50,047.80**, significantly lower than Linear Regression (**70,384.41**) and Ridge Regression (**70,387.21**). Additionally, Random Forest achieves an R^2 score of **0.8089**, meaning it explains approximately 81% of the variance in house prices. The closeness between its CV RMSE (50,664.35) and Test RMSE (50,047.80) indicates strong generalization and minimal overfitting. In contrast, both Linear and Ridge Regression show higher error and lower explanatory power (~62% R^2), demonstrating that they are unable to capture the non-linear and interaction effects present in the data. Thus, the Random Forest model is chosen as the final model for deployment and prediction.